Lab 9

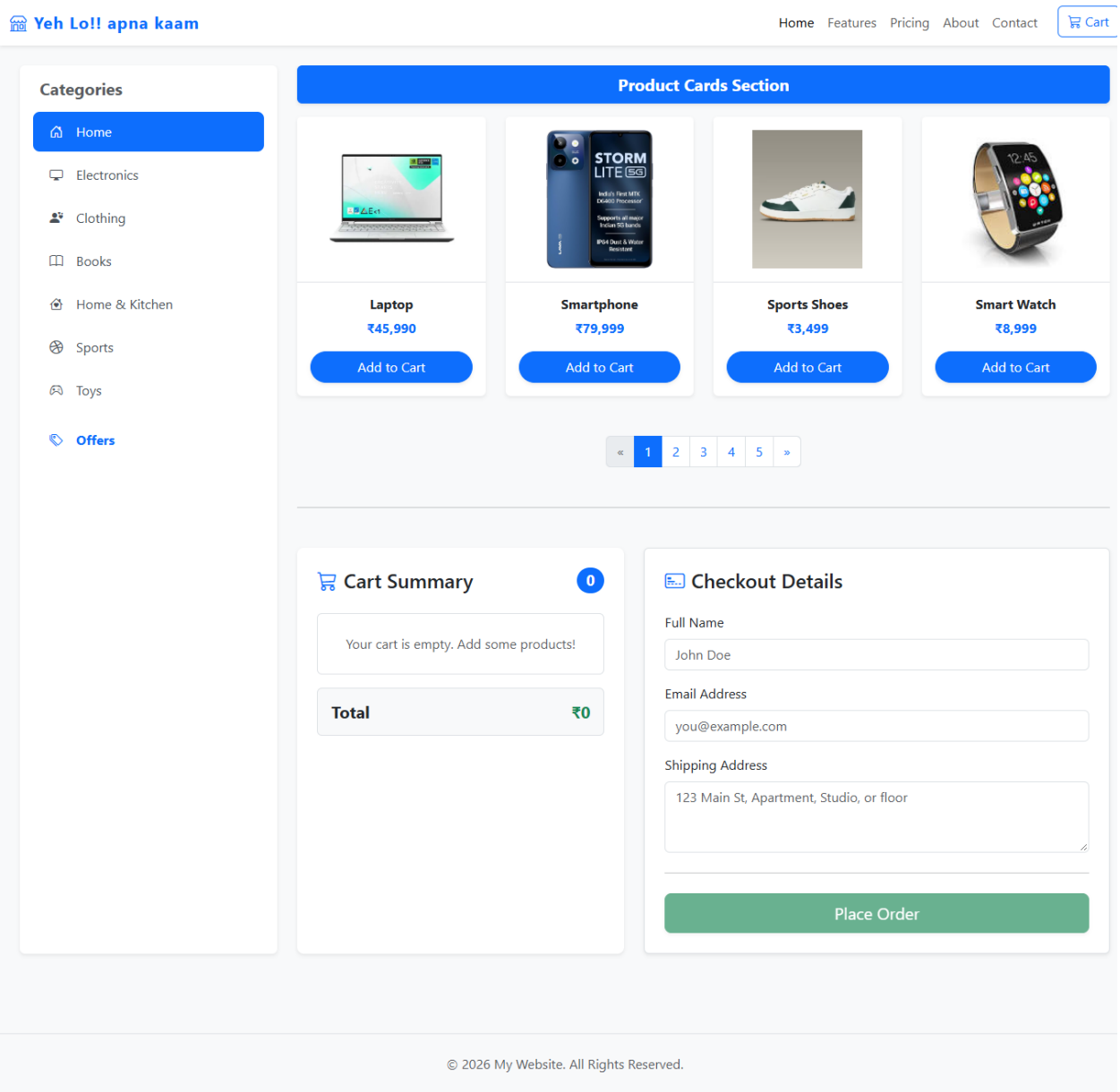
Q : Create an interactive product listing card which consist of Product image, Product Name and price and Add to Cart button and Checkout form using Bootstrap.

TAGS USED	ATTRIBUTES USED	DESCRIPTION
<!DOCTYPE html>	--	Declares the document as an HTML5 document.
<html>	lang	Defines the root element of the HTML document.
<head>	--	Contains metadata, title and stylesheet information.
<meta>	charset, name, content	Specifies character encoding and responsive viewport settings.
<title>	--	Defines the title displayed in the browser tab.
<link>	href, rel	Links the Bootstrap CSS stylesheet from the CDN.
<body>	class	Contains all visible webpage content and uses Bootstrap background styling.
<header>	class	Defines the header section and uses Bootstrap background,

		text, alignment and padding classes.
——–<img>	src, alt, class, style	Displays the Marwadi University logo and provides alternative text.
<h1>	class	Defines the main heading and uses Bootstrap font-weight styling.
<h2>	class	Defines the department heading and uses Bootstrap heading styling.
<nav>	class	Defines the navigation bar and uses Bootstrap background, alignment and padding classes.
<a>	href, class	Creates navigation links to different sections and applies Bootstrap link styling and spacing.
<section>	id, class	Defines separate webpage sections and uses Bootstrap container and margin classes.
	class	Creates Bootstrap containers, cards, card bodies, rows, columns and alert components.
<h3>	class	Defines section

		headings and uses Bootstrap text-color and font-weight classes.
<p>	class	Defines paragraphs and uses Bootstrap spacing classes where specified.
	class	Creates an unordered list and uses Bootstrap list-group styling.
	class	Creates an ordered list and uses Bootstrap numbered list-group styling.
	class	Defines list items and uses Bootstrap list-group-item styling.
<footer>	class	Defines the footer and uses Bootstrap background, text color, alignment and padding classes.

Output :


Yeh Lo!! apna kaam
Home
Features
Pricing
About
Contact
Cart
Categories
Home
Electronics
Clothing
Books
Home & Kitchen
Sports
Toys
Offers
Product Cards Section
Laptop
₹45,990
Add to Cart
Smartphone
₹79,999
Add to Cart
Sports Shoes
₹3,499
Add to Cart
Smart Watch
₹8,999
Add to Cart
« 1 2 3 4 5 »
Cart Summary
0
Your cart is empty. Add some products!
Total
₹0
Checkout Details
Full Name
John Doe
Email Address
you@example.com
Shipping Address
123 Main St, Apartment, Studio, or floor
Place Order
© 2026 My Website. All Rights Reserved.

Code :

```
<!DOCTYPE html>
<html lang="en">

<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>E-Commerce Product Listing & Checkout</title>
   <!-- Bootstrap 5 CSS -->
   <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">
   <!-- Bootstrap Icons -->
   <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-
icons@1.11.1/font/bootstrap-icons.css">
   <style>
      body {
         background-color: #f8f9fa;
         font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      }

      .card-img-top {
         height: 200px;
         object-fit: contain;
         padding: 1rem;
         background-color: #fff;
      }

      .sidebar-menu .list-group-item {
         border: none;
         padding: 0.75rem 1.25rem;
```

```
    margin-bottom: 0.25rem;
    border-radius: 0.5rem;
    color: #495057;
    transition: all 0.2s ease;
}

.sidebar-menu .list-group-item:hover {
    background-color: #e9ecef;
}

.sidebar-menu .list-group-item.active {
    background-color: #0d6efd;
    color: white;
}

.navbar-brand {
    font-weight: bold;
    letter-spacing: 1px;
}

.cart-summary {
    background-color: #fff;
    border-radius: 0.5rem;
    box-shadow: 0 0.125rem 0.25rem rgba(0, 0, 0, 0.075);
}

/* Custom Notification Toast Position */
.toast-container {
    position: fixed;
    bottom: 20px;
    right: 20px;
    z-index: 1055;
}
```

```
  </style>
</head>

<body>

  <!-- Header / Navbar -->
  <nav class="navbar navbar-expand-lg navbar-light bg-white shadow-sm mb-4">
    <div class="container-fluid">
      <a class="navbar-brand text-primary" href="#">
        <i class="bi bi-shop me-2"></i>Yeh Lo!! apna kaam
      </a>
      <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
data-bs-toggle="collapse"
        data-bs-target="#navbarNav">
        <span class="navbar-toggler-icon"></span>
      </button>
      <div class="collapse navbar-collapse justify-content-end" id="navbarNav">
        <ul class="navbar-nav">
          <li class="nav-item"><a class="nav-link active"
href="#">Home</a></li>
          <li class="nav-item"><a class="nav-link" href="#">Features</a></li>
          <li class="nav-item"><a class="nav-link" href="#">Pricing</a></li>
          <li class="nav-item"><a class="nav-link" href="#">About</a></li>
          <li class="nav-item"><a class="nav-link" href="#">Contact</a></li>
          <li class="nav-item ms-lg-3">
            <button class="btn btn-outline-primary position-relative"
onclick="scrollToCart()">
              <i class="bi bi-cart3"></i> Cart
              <span id="cart-badge"
                class="position-absolute top-0 start-100 translate-middle badge
rounded-pill bg-danger d-none">
                0
              </span>
```

```
                </button>
              </li>
            </ul>
          </div>
        </div>
      </nav>

      <!-- Main Content Container -->
      <div class="container-fluid px-4 pb-5">
        <div class="row">

          <!-- Sidebar -->
          <div class="col-md-3 mb-4">
            <div class="bg-white p-3 rounded shadow-sm h-100 sidebar-menu">
              <h5 class="mb-3 ps-2 text-muted fw-bold">Categories</h5>
              <div class="list-group">
                <a href="#" class="list-group-item list-group-item-action active d-
flex align-items-center">
                  <i class="bi bi-house-door me-3"></i> Home
                </a>
                <a href="#" class="list-group-item list-group-item-action d-flex align-
items-center">
                  <i class="bi bi-display me-3"></i> Electronics
                </a>
                <a href="#" class="list-group-item list-group-item-action d-flex align-
items-center">
                  <i class="bi bi-person-hearts me-3"></i> Clothing
                </a>
                <a href="#" class="list-group-item list-group-item-action d-flex align-
items-center">
                  <i class="bi bi-book me-3"></i> Books
                </a>
```

```
                <a href="#" class="list-group-item list-group-item-action d-flex align-items-center">
                    <i class="bi bi-house-heart me-3"></i> Home & Kitchen
                </a>
                <a href="#" class="list-group-item list-group-item-action d-flex align-items-center">
                    <i class="bi bi-dribbble me-3"></i> Sports
                </a>
                <a href="#" class="list-group-item list-group-item-action d-flex align-items-center">
                    <i class="bi bi-controller me-3"></i> Toys
                </a>
                <a href="#"
                    class="list-group-item list-group-item-action d-flex align-items-center text-primary fw-bold mt-2">
                    <i class="bi bi-tags me-3"></i> Offers
                </a>
            </div>
        </div>
    </div>

    <!-- Main Content Area -->
    <div class="col-md-9">

        <!-- Section Title -->
        <div class="bg-primary text-white p-2 rounded mb-3 text-center fw-bold fs-5 shadow-sm">
            Product Cards Section
        </div>

        <!-- Product Grid -->
        <div class="row mb-4">
            <!-- Product 1 -->
```

```
            <div class="col-12 col-sm-6 col-lg-3 mb-4">
              <div class="card h-100 shadow-sm border-0 text-center">
                <!-- Image fallback script included directly inline to handle missing
local files elegantly -->
                <img src="laptop.jpg" class="card-img-top border-bottom"
alt="Laptop"
                  onerror="this.onerror=null;this.src='https://placehold.co/400x3
00/e9ecef/495057?text=Laptop';">
                <div class="card-body d-flex flex-column">
                  <h6 class="card-title fw-bold text-dark">Laptop</h6>
                  <p class="card-text text-primary fw-bold mb-3">₹45,990</p>
                  <button class="btn btn-primary mt-auto rounded-pill w-100
shadow-sm"
                    onclick="addToCart('Laptop', 45990)">
                    Add to Cart
                  </button>
                </div>
              </div>
            </div>

            <!-- Product 2 -->
            <div class="col-12 col-sm-6 col-lg-3 mb-4">
              <div class="card h-100 shadow-sm border-0 text-center">
                <img src="smartphone.jpg" class="card-img-top border-bottom"
alt="Smartphone"
                  onerror="this.onerror=null;this.src='https://placehold.co/400x3
00/e9ecef/495057?text=Smartphone';">
                <div class="card-body d-flex flex-column">
                  <h6 class="card-title fw-bold text-dark">Smartphone</h6>
                  <p class="card-text text-primary fw-bold mb-3">₹79,999</p>
                  <button class="btn btn-primary mt-auto rounded-pill w-100
shadow-sm"
                    onclick="addToCart('Smartphone', 79999)">
```

```
                    Add to Cart
                  </button>
                </div>
              </div>
            </div>

            <!-- Product 3 -->
            <div class="col-12 col-sm-6 col-lg-3 mb-4">
              <div class="card h-100 shadow-sm border-0 text-center">
                <img src="sports_shoes.jpg" class="card-img-top border-bottom"
alt="Sports Shoes"
                  onerror="this.onerror=null;this.src='https://placehold.co/400x3
00/e9ecef/495057?text=Sports+Shoes';">
                <div class="card-body d-flex flex-column">
                  <h6 class="card-title fw-bold text-dark">Sports Shoes</h6>
                  <p class="card-text text-primary fw-bold mb-3">₹3,499</p>
                  <button class="btn btn-primary mt-auto rounded-pill w-100
shadow-sm"
                    onclick="addToCart('Sports Shoes', 3499)">
                    Add to Cart
                  </button>
                </div>
              </div>
            </div>

            <!-- Product 4 -->
            <div class="col-12 col-sm-6 col-lg-3 mb-4">
              <div class="card h-100 shadow-sm border-0 text-center">
                <img src="smartwatch.jpg" class="card-img-top border-bottom"
alt="Smart Watch"
                  onerror="this.onerror=null;this.src='https://placehold.co/400x3
00/e9ecef/495057?text=Smart+Watch';">
                <div class="card-body d-flex flex-column">
```

```
                    <h6 class="card-title fw-bold text-dark">Smart Watch</h6>
                    <p class="card-text text-primary fw-bold mb-3">₹8,999</p>
                    <button class="btn btn-primary mt-auto rounded-pill w-100
shadow-sm"
                      onclick="addToCart('Smart Watch', 8999)">
                      Add to Cart
                    </button>
                  </div>
                </div>
              </div>
            </div>

            <!-- Pagination (As per wireframe) -->
            <nav aria-label="Product pagination" class="mb-5">
              <ul class="pagination justify-content-center">
                <li class="page-item disabled"><a class="page-link" href="#" aria-
label="Previous"><span
                      aria-hidden="true">&laquo;</span></a></li>
                <li class="page-item active"><a class="page-link"
href="#">1</a></li>
                <li class="page-item"><a class="page-link" href="#">2</a></li>
                <li class="page-item"><a class="page-link" href="#">3</a></li>
                <li class="page-item"><a class="page-link" href="#">4</a></li>
                <li class="page-item"><a class="page-link" href="#">5</a></li>
                <li class="page-item"><a class="page-link" href="#" aria-
label="Next"><span
                      aria-hidden="true">&raquo;</span></a></li>
              </ul>
            </nav>

            <hr class="mb-5">

            <!-- Cart and Checkout Section -->
```

```
        <div class="row" id="checkout-section">

          <!-- Cart Summary -->
          <div class="col-lg-5 mb-4">
            <div class="cart-summary p-4 h-100">
              <h4 class="mb-4 d-flex justify-content-between align-items-
center">
                <span><i class="bi bi-cart3 text-primary me-2"></i>Cart
Summary</span>
                <span id="cart-count" class="badge bg-primary rounded-
pill">0</span>
              </h4>

              <ul class="list-group mb-3" id="cart-items">
                <!-- Cart items will be dynamically injected here -->
                <li class="list-group-item text-center text-muted py-4"
id="empty-cart-msg">
                  Your cart is empty. Add some products!
                </li>
              </ul>

              <div class="d-flex justify-content-between align-items-center bg-
light p-3 rounded border">
                <h5 class="mb-0 fw-bold">Total</h5>
                <h5 class="mb-0 fw-bold text-success" id="cart-total">₹0</h5>
              </div>
            </div>
          </div>

          <!-- Checkout Form -->
          <div class="col-lg-7 mb-4">
            <div class="bg-white p-4 rounded shadow-sm border h-100">
```

```
                <h4 class="mb-4"><i class="bi bi-credit-card-2-front text-primary
me-2"></i>Checkout Details
                </h4>

                <!-- Success Message Container (Hidden by default) -->
                <div id="order-success-msg" class="alert alert-success d-none
shadow-sm" role="alert">
                   <i class="bi bi-check-circle-fill me-2"></i>
                   <strong>Order Placed Successfully!</strong> Thank you for your
purchase.
                </div>

                <form id="checkout-form" onsubmit="handleCheckout(event)">
                   <div class="row g-3">
                      <div class="col-sm-12">
                         <label for="fullName" class="form-label">Full
Name</label>
                         <input type="text" class="form-control" id="fullName"
placeholder="John Doe"
                            required>
                      </div>
                      <div class="col-sm-12">
                         <label for="email" class="form-label">Email
Address</label>
                         <input type="email" class="form-control" id="email"
                            placeholder="you@example.com" required>
                      </div>
                      <div class="col-12">
                         <label for="address" class="form-label">Shipping
Address</label>
                         <textarea class="form-control" id="address" rows="3"
                            placeholder="123 Main St, Apartment, Studio, or floor"
required></textarea>
```

```
                    </div>
                  </div>
                  <hr class="my-4">
                  <button class="btn btn-success w-100 btn-lg shadow-sm"
type="submit"
                    id="place-order-btn" disabled>
                    Place Order
                  </button>
                </form>
              </div>
            </div>

          </div>
        </div>
      </div>
    </div>

    <!-- Footer -->
    <footer class="bg-light text-center text-muted py-4 border-top mt-4">
      <div class="container">
        <p class="mb-0">© 2026 My Website. All Rights Reserved.</p>
      </div>
    </footer>

    <!-- Toast Notification for Add to Cart -->
    <div class="toast-container">
      <div id="cartToast" class="toast align-items-center text-bg-primary border-0"
role="alert" aria-live="assertive"
        aria-atomic="true">
        <div class="d-flex">
          <div class="toast-body" id="toast-msg">
            Item added to cart!
          </div>
```

```
            <button type="button" class="btn-close btn-close-white me-2 m-auto"
data-bs-dismiss="toast"
                aria-label="Close"></button>
        </div>
    </div>
</div>

<!-- Bootstrap 5 JS Bundle with Popper -->
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.
js"></script>

<!-- Custom JavaScript -->
<script>
    // State management
    let cart = [];
    let total = 0;

    // Currency Formatter (Indian Rupees)
    const formatCurrency = (amount) => {
        return new Intl.NumberFormat('en-IN', {
            style: 'currency',
            currency: 'INR',
            maximumFractionDigits: 0
        }).format(amount);
    };

    // Add to Cart Logic
    function addToCart(name, price) {
        // Check if item already exists in cart
        const existingItem = cart.find(item => item.name === name);
        if (existingItem) {
            existingItem.quantity += 1;
```

```
        } else {
            cart.push({ name: name, price: price, quantity: 1 });
        }

        updateCartUI();
        showToast(`${name} added to cart!`);
    }

    // Remove from Cart Logic
    function removeFromCart(index) {
        cart.splice(index, 1);
        updateCartUI();
    }

    // Update UI Logic
    function updateCartUI() {
        const cartItemsList = document.getElementById('cart-items');
        const cartCountElement = document.getElementById('cart-count');
        const cartBadgeElement = document.getElementById('cart-badge');
        const cartTotalElement = document.getElementById('cart-total');
        const emptyCartMsg = document.getElementById('empty-cart-msg');
        const placeOrderBtn = document.getElementById('place-order-btn');

        // Clear current list (keep the empty message template hidden for a
moment)
        cartItemsList.innerHTML = '';
        total = 0;
        let totalItems = 0;

        if (cart.length === 0) {
            cartItemsList.innerHTML = `<li class="list-group-item text-center text-
muted py-4" id="empty-cart-msg">Your cart is empty. Add some products!</li>`;
            placeOrderBtn.disabled = true;
```

```
            cartBadgeElement.classList.add('d-none');
        } else {
            placeOrderBtn.disabled = false;
            cartBadgeElement.classList.remove('d-none');

            // Re-build cart list
            cart.forEach((item, index) => {
                const itemTotal = item.price * item.quantity;
                total += itemTotal;
                totalItems += item.quantity;

                const li = document.createElement('li');
                li.className = 'list-group-item d-flex justify-content-between lh-sm py-
3';
                li.innerHTML = `
                    <div>
                        <h6 class="my-0">${item.name}</h6>
                        <small class="text-muted">Qty: ${item.quantity} x
${formatCurrency(item.price)}</small>
                    </div>
                    <div class="text-end d-flex flex-column justify-content-between
align-items-end">
                        <span class="text-muted mb-
2">${formatCurrency(itemTotal)}</span>
                        <button class="btn btn-sm btn-outline-danger" style="font-size:
0.7rem; padding: 0.1rem 0.4rem;" onclick="removeFromCart(${index})"><i
class="bi bi-trash"></i></button>
                    </div>
                `;
                cartItemsList.appendChild(li);
            });
        }
```

```
        // Update Counts and Totals
        cartCountElement.textContent = totalItems;
        cartBadgeElement.textContent = totalItems;
        cartTotalElement.textContent = formatCurrency(total);
    }

    // Scroll to cart helper
    function scrollToCart() {
        document.getElementById('checkout-section').scrollIntoView({ behavior:
'smooth' });
    }

    // Show Bootstrap Toast
    function showToast(message) {
        const toastEl = document.getElementById('cartToast');
        document.getElementById('toast-msg').textContent = message;
        const toast = new bootstrap.Toast(toastEl, { delay: 3000 });
        toast.show();
    }

    // Handle Checkout Form Submission
    function handleCheckout(event) {
        event.preventDefault(); // Prevent page reload

        if (cart.length === 0) return;

        // Hide form, show success message
        document.getElementById('checkout-form').classList.add('d-none');
        document.getElementById('order-success-msg').classList.remove('d-none');

        // Empty cart
        cart = [];
        updateCartUI();
```

```
        // Reset form for future (optional delay to reset visual state)
        setTimeout(() => {
            document.getElementById('checkout-form').reset();
            document.getElementById('checkout-form').classList.remove('d-none');
            document.getElementById('order-success-msg').classList.add('d-none');
            window.scrollTo({ top: 0, behavior: 'smooth' });
        }, 4000);
    }
  </script>
</body>

</html>
```